

# Scaling deep learning for materials discovery

<https://doi.org/10.1038/s41586-023-06735-9>

Received: 8 May 2023

Accepted: 10 October 2023

Published online: 29 November 2023

Open access

 Check for updates

 Amil Merchant<sup>1,3</sup>, Simon Batzner<sup>1,3</sup>, Samuel S. Schoenholz<sup>1,3</sup>, Muratahan Aykol<sup>1</sup>, Gwoon Cheon<sup>2</sup> & Ekin Dogus Cubuk<sup>1,3</sup>

Novel functional materials enable fundamental breakthroughs across technological applications from clean energy to information processing<sup>1–11</sup>. From microchips to batteries and photovoltaics, discovery of inorganic crystals has been bottlenecked by expensive trial-and-error approaches. Concurrently, deep-learning models for language, vision and biology have showcased emergent predictive capabilities with increasing data and computation<sup>12–14</sup>. Here we show that graph networks trained at scale can reach unprecedented levels of generalization, improving the efficiency of materials discovery by an order of magnitude. Building on 48,000 stable crystals identified in continuing studies<sup>15–17</sup>, improved efficiency enables the discovery of 2.2 million structures below the current convex hull, many of which escaped previous human chemical intuition. Our work represents an order-of-magnitude expansion in stable materials known to humanity. Stable discoveries that are on the final convex hull will be made available to screen for technological applications, as we demonstrate for layered materials and solid-electrolyte candidates. Of the stable structures, 736 have already been independently experimentally realized. The scale and diversity of hundreds of millions of first-principles calculations also unlock modelling capabilities for downstream applications, leading in particular to highly accurate and robust learned interatomic potentials that can be used in condensed-phase molecular-dynamics simulations and high-fidelity zero-shot prediction of ionic conductivity.

The discovery of energetically favourable inorganic crystals is of fundamental scientific and technological interest in solid-state chemistry. Experimental approaches over the decades have catalogued 20,000 computationally stable structures (out of a total of 200,000 entries) in the Inorganic Crystal Structure Database (ICSD)<sup>15,18</sup>. However, this strategy is impractical to scale owing to costs, throughput and synthesis complications<sup>19</sup>. Instead, computational approaches championed by the Materials Project (MP)<sup>16</sup>, the Open Quantum Materials Database (OQMD)<sup>17</sup>, AFLOWLIB<sup>20</sup> and NOMAD<sup>21</sup> have used first-principles calculations based on density functional theory (DFT) as approximations of physical energies. Combining *ab initio* calculations with simple substitutions has allowed researchers to improve to 48,000 computationally stable materials according to our own recalculations<sup>22–24</sup> (see Methods). Although data-driven methods that aid in further materials discovery have been pursued, thus far, machine-learning techniques have been ineffective in estimating stability (decomposition energy) with respect to the convex hull of energies from competing phases<sup>25</sup>.

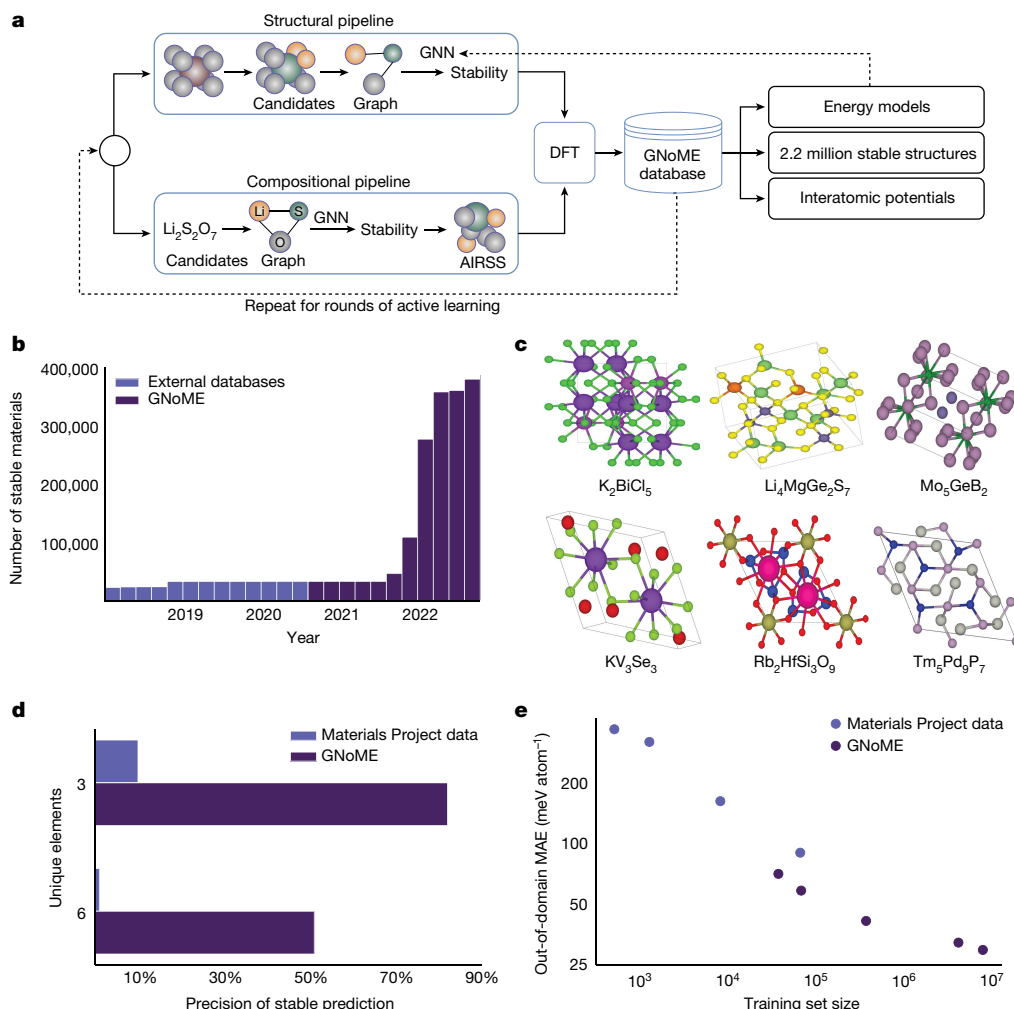
In this paper, we scale up machine learning for materials exploration through large-scale active learning, yielding the first models that accurately predict stability and, therefore, can guide materials discovery. Our approach relies on two pillars: first, we establish methods for generating diverse candidate structures, including new symmetry-aware partial substitutions (SAPS) and random structure search<sup>26</sup>. Second, we use state-of-the-art graph neural networks (GNNs) that improve modelling of material properties given structure or composition. In a series of rounds, these graph networks for materials exploration (GNoME) are trained on available data and used to filter candidate structures.

The energy of the filtered candidates is computed using DFT, both verifying model predictions and serving as a data flywheel to train more robust models on larger datasets in the next round of active learning.

Through this iterative procedure, GNoME models have discovered more than 2.2 million structures stable with respect to previous work, in particular agglomerated datasets encompassing computational and experimental structures<sup>15–17,27</sup>. Given that discovered materials compete for stability, the updated convex hull consists of 381,000 new entries for a total of 421,000 stable crystals, representing an order-of-magnitude expansion from all previous discoveries. Consistent with observations in other domains of machine learning<sup>28</sup>, we observe that our neural networks predictions improve as a power law with the amount of data. Final GNoME models accurately predict energies to 11 meV atom<sup>−1</sup> and improve the precision of stable predictions (hit rate) to above 80% with structure and 33% per 100 trials with composition only, compared with 1% in previous work<sup>17</sup>. Moreover, these networks develop emergent out-of-distribution generalization. For example, GNoME enables accurate predictions of structures with 5+ unique elements (despite omission from training), providing one of the first strategies to efficiently explore this chemical space. We validate findings by comparing predictions with experiments and higher-fidelity *r*<sup>2</sup>SCAN (ref. 29) computations.

Finally, we demonstrate that the dataset produced in GNoME discovery unlocks new modelling capabilities for downstream applications. The structures and relaxation trajectories present a large and diverse dataset to enable training of learned, equivariant interatomic potentials<sup>30,31</sup> with unprecedented accuracy and zero-shot generalization.

<sup>1</sup>Google DeepMind, Mountain View, CA, USA. <sup>2</sup>Google Research, Mountain View, CA, USA. <sup>3</sup>These authors contributed equally: Amil Merchant, Simon Batzner, Samuel S. Schoenholz, Ekin Dogus Cubuk. ✉e-mail: amilmerchant@google.com; cubuk@google.com



**Fig. 1 | GNoME enables efficient discovery.** **a**, A summary of the GNoME-based discovery shows how model-based filtration and DFT serve as a data flywheel to improve predictions. **b**, Exploration enabled by GNoME has led to 381,000 new stable materials, almost an order of magnitude larger than previous work. **c**, 736 structures have been independently experimentally verified, with six examples shown<sup>50–55</sup>. **d**, Improvements from graph network predictions enable

efficient discovery in combinatorial regions of materials, for example, with six unique elements, even though the training set stopped at four unique elements. **e**, GNoME showcases emergent generalization when tested on out-of-domain inputs from random structure search, indicating progress towards a universal energy model.

We demonstrate the promise of these potentials for materials property prediction through the estimation of ionic conductivity from molecular-dynamics simulations.

## Overview of generation and filtration

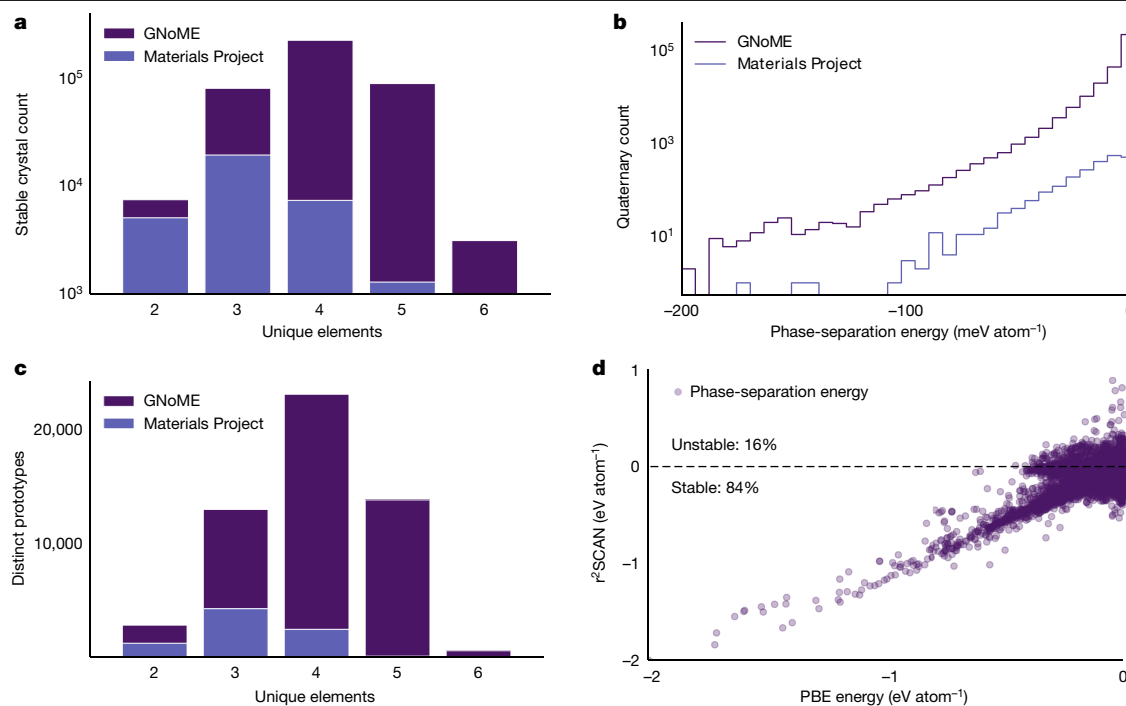
The space of possible materials is far too large to sample in an unbiased manner. Without a reliable model to cheaply approximate the energy of candidates, researchers guided searches by restricting generation with chemical intuition, accomplished by substituting similar ions or enumerating prototypes<sup>22</sup>. Although improving search efficiency<sup>17,27</sup>, this strategy fundamentally limited how diverse candidates could be. By guiding searches with neural networks, we are able to use diversified methods for generating candidates and perform a broader exploration of crystal space without sacrificing efficiency.

To generate and filter candidates, we use two frameworks, which are visualized in Fig. 1a. First, structural candidates are generated by modifications of available crystals. However, we strongly augment the set of substitutions by adjusting ionic substitution probabilities to give priority to discovery and use newly proposed symmetry aware partial substitutions (SAPS) to efficiently enable incomplete replacements<sup>32</sup>. This expansion results in more than 10<sup>9</sup> candidates over the

course of active learning; the resulting structures are filtered by means of GNoME using volume-based test-time augmentation and uncertainty quantification through deep ensembles<sup>33</sup>. Finally, structures are clustered and polymorphs are ranked for evaluation with DFT (see Methods). In the second framework, compositional models predict stability without structural information. Inputs are reduced chemical formulas. Generation by means of oxidation-state balancing is often too strict (for example, neglecting  $\text{Li}_{15}\text{Si}_4$ ). Using relaxed constraints (see Methods), we filter compositions using GNoME and initialize 100 random structures for evaluation through ab initio random structure searching (AIRSS)<sup>26</sup>. In both frameworks, models provide a prediction of energy and a threshold is chosen on the basis of the relative stability (decomposition energy) with respect to competing phases. Evaluation is performed through DFT computations in the Vienna Ab initio Simulation Package (VASP)<sup>34</sup> and we measure both the number of stable materials discovered as well as the precision of predicted stable materials (hit rate) in comparison with the Materials Project<sup>16</sup>.

## GNoME

All GNoME models are GNNs that predict the total energy of a crystal. Inputs are converted to a graph through a one-hot embedding of the



**Fig. 2 | Summaries of discovered stable crystals.** **a**, GNoME enables efficient discovery in the combinatorial spaces of 4+ unique elements that can be difficult for human experts. **b**, Phase-separation energies (energy to the convex hull) for discovered quaternaries showcase similar patterns but larger absolute numbers

elements. We follow the message-passing formulation<sup>35,36</sup>, in which aggregate projections are shallow multilayer perceptrons (MLPs) with swish nonlinearities. For structural models, we find it important to normalize messages from edges to nodes by the average adjacency of atoms across the entire dataset. Initial models are trained on a snapshot of the Materials Project from 2018 of approximately 69,000 materials. Previous work benchmarked this task at a mean absolute error (MAE) of 28 meV atom<sup>-1</sup> (ref. 37); however, we find that the improved networks achieve a MAE of 21 meV atom<sup>-1</sup>. We fix this promising architecture (see Methods) and focus on scaling in the rest of this paper.

### Active learning

A core step in our framework for accelerating materials discovery is active learning. In both structural and compositional frameworks, candidate structures filtered using GNoME are evaluated using DFT calculations with standardized settings from the Materials Project. Resulting energies of relaxed structures not only verify the stability of crystal structures but are also incorporated into the iterative active-learning workflow as further training data and structures for candidate generation. Whereas the hit rate for both structural and compositional frameworks start at less than 6% and 3%, respectively, performance improves steadily through six rounds of active learning. Final ensembles of GNoME models improve to a prediction error of 11 meV atom<sup>-1</sup> on relaxed structures and hit rates of greater than 80% and 33%, respectively, clearly showing the benefits of scale. An analysis of final GNoME hit rates is provided in Fig. 1d.

### Scaling laws and generalization

The test loss performance of GNoME models exhibit improvement as a power law with further data. These results are in line with neural scaling laws in deep learning<sup>28,38</sup> and suggest that further discovery efforts could continue to improve generalization. Emphatically, unlike the case of language or vision, in materials science, we can continue to generate data and discover stable crystals, which can be reused to continue

than previous catalogues. **c**, Discovered stable crystals correspond to 45,500 novel prototypes as measured by XtalFinder (ref. 39). **d**, Validation by r<sup>2</sup>SCAN shows that 84% of discovered binary and ternary crystals retain negative phase separations with more accurate functionals.

scaling up the model. We also demonstrate emergent generalization to out-of-distribution tasks by testing structural models trained on data originating from substitutions on crystals arising from random search<sup>26</sup> in Fig. 1e. These examples are often high-energy local minima and out of distribution compared with data generated by our structural pipeline (which, by virtue of substitutions, contains structures near their minima). Nonetheless, we observe clear improvement with scale. These results indicate that final GNoME models are a substantial step towards providing the community with a universal energy predictor, capable of handling diverse materials structures through deep learning.

### Discovered stable crystals

Using the described process of scaling deep learning for materials exploration, we increase the number of known stable crystals by almost an order of magnitude. In particular, GNoME models found 2.2 million crystal structures stable with respect to the Materials Project. Of these, 381,000 entries live on the updated convex hull as newly discovered materials.

Consistent with other literature on structure prediction, the GNoME materials could be bumped off the convex hull by future discoveries, similar to how GNoME displaces at least 5,000 ‘stable’ materials from the Materials Project and the OQMD. See Supplementary Note 1 for discussion on improving structures of already-discovered compositions. Nevertheless, Figs. 1 and 2 provide a summary of the stable materials, with Fig. 1b focusing on the growth over time. We see substantial gains in the number of structures with more than four unique elements in Fig. 2a. This is particularly promising because these materials have proved difficult for previous discovery efforts<sup>27</sup>. Our scaled GNoME models overcome this obstacle and enable efficient discovery in combinatorially large regions.

Clustering by means of prototype analysis<sup>39</sup> supports the diversity of discovered crystals with GNoME, leading to more than 45,500 novel

prototypes in Fig. 2c (a 5.6 times increase from 8,000 of the Materials Project), which could not have arisen from full substitutions or prototype enumeration. Finally, in Fig. 2b, we compare the phase-separation energy (also referred to as the decomposition enthalpy) of discovered quaternaries with those from the Materials Project to measure the relative distance to the convex hull of all other competing phases. The similarities in distribution suggest that the found materials are meaningfully stable with respect to competing phases and not just ‘filling in the convex hull.’ Further analyses of materials near to (but not on) the updated convex hull is given in Supplementary Note 3.

### Validation through experimental matching and $r^2$ SCAN

All candidates for GNoME are derived from snapshots of databases made in March 2021, including the Materials Project and the OQMD. Concurrent to our discovery efforts, researchers have continued to experimentally create new crystals, providing a way to validate GNoME findings. Of the experimental structures aggregated in the ICSD, 736 match structures that were independently obtained through GNoME. Six of the experimentally matched structures are presented in Fig. 1c and further details of the experimental matches are provided in Supplementary Note 1. Similarly, of the 3,182 compositions added to the Materials Project since the snapshot, 2,202 are available in the GNoME database and 91% match on structure. A manual check of ‘newly’ discovered crystals supported the findings, with details in Supplementary Note 4.

We also validate predictions to ensure that model-based exploration did not overfit simulation parameters. We focus on the choice of functional. Standard projector augmented wave (PAW)-Perdew-Burke-Ernzerhof (PBE) potentials provided a speed-accuracy trade-off suited for large-scale discovery<sup>40,41</sup>, but the  $r^2$ SCAN functional provides a more accurate meta-generalized gradient approximation<sup>29,42,43</sup>. 84% of the discovered binaries and ternary materials also present negative phase-separation energies (as visualized in Fig. 2d, comparable with a 90% ratio in the Materials Project but operating at a larger scale). 86.8% of tested quaternaries also remain stable on the  $r^2$ SCAN convex hull. The discrepancies between PBE and  $r^2$ SCAN energies are further analysed in Supplementary Note 2.

### Composition families of interest

We highlight the benefits of a catalogue of stable materials an order of magnitude larger than previous work. When searching for a material with certain desirable properties, researchers often filter such catalogues, as computational stability is often linked with experimental realizability. We perform similar analyses for three applications. First, layered materials are promising systems for electronics and energy storage<sup>44</sup>. Methods from previous studies<sup>45</sup> suggest that approximately 1,000 layered materials are stable compared with the Materials Project, whereas this number increases to about 52,000 with GNoME-based discoveries. Similarly, following a holistic screening approach with filters such as exclusion of transition metals or by lithium fraction, we find 528 promising Li-ion conductors among GNoME discoveries, a 25 times increase compared with the original study<sup>46</sup>. Finally, Li/Mn transition-metal oxides are a promising family to replace LiCoO<sub>2</sub> in rechargeable batteries<sup>25</sup> and GNoME has discovered an extra 15 candidates stable relative to the Materials Project compared with the original nine.

### Scaling up learned interatomic potentials

The process of discovery of stable crystals also provides a data source beyond stable materials. In particular, the ionic relaxations involve computation of first-principles energies and forces for a diverse set of materials structures. This generates a dataset of unprecedented diversity and scale, which we explore to pretrain a general-purpose

machine-learning interatomic potential (MLIP) for bulk solids. MLIPs have become a promising tool to accelerate the simulation of materials by learning the energies and forces of reference structures computed at first-principles accuracy<sup>30,47–49</sup>. Existing efforts typically train models per material, with data often sampled from ab initio molecular dynamics (AIMD). This markedly limits their general applicability and adoption, requiring expensive data collection and training a new potential from scratch for each system. By making use of the GNoME dataset of first-principles calculations from diverse structural relaxations, we demonstrate that large-scale pretraining of MLIPs enables models that show unprecedented zero-shot accuracy and can be used to discover superionic conductors, without training on any material-specific data.

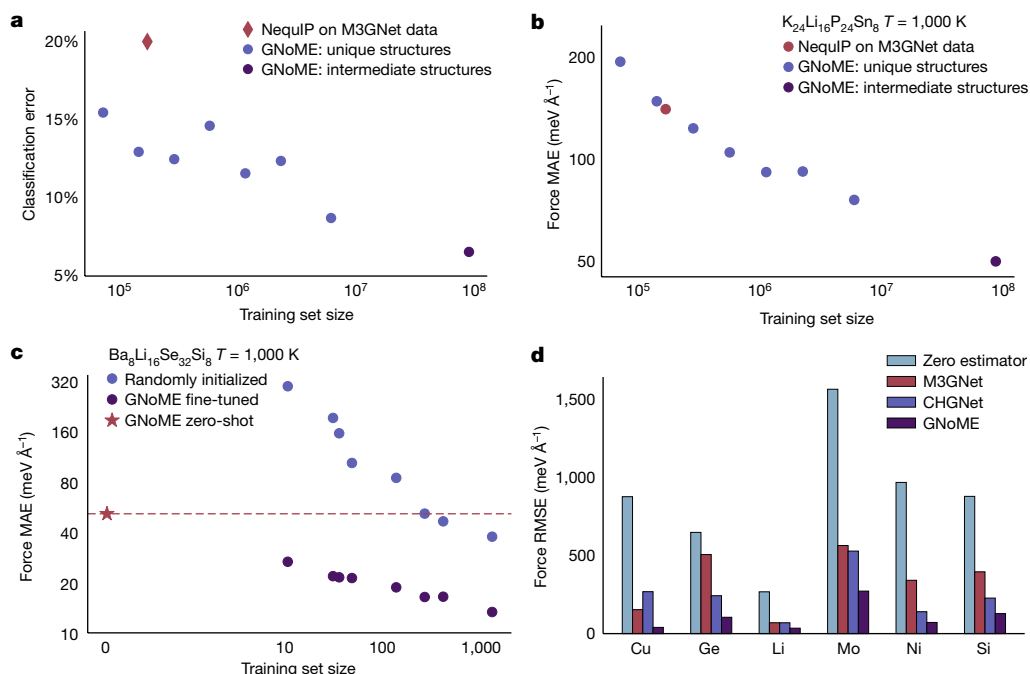
### Zero-shot scaling and generalization

We scale pretraining of a NequIP potential<sup>30</sup> on data sampled from ionic relaxations. Increasing the pretraining dataset, we observe consistent power-law improvements in accuracy (see Fig. 3a,b). Despite only being trained on ionic relaxations and not on molecular-dynamics data, the pretrained GNoME potential shows remarkable accuracy when evaluated on downstream data sampled from the new distribution of AIMD in a zero-shot manner, that is, in which no training data originate from AIMD simulations (see Fig. 3). Notably, this includes unseen compositions, melted structures and structures including vacancies, all of which are not included in our training set (see Supplementary Note 6.4). In particular, we find that the scale of the GNoME dataset allows it to outperform existing general-purpose potentials (see Fig. 3d) and makes the pretrained potential competitive with models trained explicitly on hundreds of samples from the target data distributions (see Supplementary Note 6.4). We observe particularly pronounced improvements in the transferability of MLIPs, one of the most pressing shortcomings of MLIPs. To assess the transferability of the potentials, we test their performance under distribution shift: we train two types of NequIP potential on structures sampled from AIMD at  $T = 400$  K, one in which the network is trained from randomly initialized weights and the other in which we fine-tune from a pretrained GNoME checkpoint. We then measure the performance of both potentials on data sampled from AIMD at  $T = 1,000$  K (see Fig. 3c), out of distribution with respect to the 400-K data. The potential pretrained on GNoME data shows systematic and strong improvements in transferability over the potential trained from scratch, even when training is performed on more than 1,000 structures. The zero-shot GNoME potential, not fine-tuned on any data from this composition, outperforms even a state-of-the-art NequIP model trained on hundreds of structures.

### Screening solid-state ionic conductors

Solid electrolytes are a core component of solid-state batteries, promising higher energy density and safety than liquid electrolytes, but suffer from lower ionic conductivities at present. In the search for novel electrolyte materials, AIMD allows for the prediction of ionic conductivities from first principles. However, owing to the poor scaling of DFT with the number of electrons, routine simulations are limited to hundreds of picoseconds, hundreds of atoms and, most importantly, small compositional search spaces. Here we show that the GNoME potentials show high robustness in this out-of-distribution, zero-shot setting and generalizes to high temperatures, which allows them to serve as a tool for high-throughput discovery of novel solid-state electrolytes. We use GNoME potentials pretrained on datasets of increasing size in molecular-dynamics simulations on 623 never-before-seen compositions. Figure 3a shows the ability of the pretrained GNoME potentials to classify unseen compositions as superionic conductors in comparison with AIMD.

When scaled to the GNoME dataset—much larger than existing approaches—we find that deep learning unlocks previously impossible capabilities for building transferable interatomic potentials for



**Fig. 3 | Scaling learned interatomic potentials.** **a**, Classification of whether a material is a superionic conductor as predicted by GNoME-driven simulations in comparison with AIMD, tested on 623 unseen compositions. The classification error improves as a power law with training set size. **b**, Zero-shot force error as a function of training set size for the unseen material  $K_{24}Li_{16}P_{24}Sn_8$ . **c**, Robustness under distribution shift, showing the MAE in forces on the example material  $Ba_8Li_{16}Se_{32}Si_8$ . A GNoME-pretrained and a randomly initialized potential are

trained on data of various sizes sampled at  $T = 400$  K and evaluated on data sampled at  $T = 1,000$  K. The zero-shot GNoME potential outperforms state-of-the-art models trained from scratch on hundreds of structures. **d**, Comparison of zero-shot force errors of three different pretrained, general-purpose potentials for bulk systems on the test set of ref. 56. Note that the composition Ni is not present in the GNoME pretraining data. RMSE, root-mean-square error.

inorganic bulk crystals and allows for high-accuracy, zero-shot prediction of materials properties at scale.

## Conclusion

We show that GNNs trained on a large and diverse set of first-principles calculations can enable the efficient discovery of inorganic materials, increasing the number of stable crystals by more than an order of magnitude. Associated datasets empower machine-learned interatomic potentials, giving accurate and robust molecular-dynamics simulations out of the box on unseen bulk materials. Our findings raise interesting questions about the capabilities of deep-learning systems in the natural sciences: the application of machine-learning methods for scientific discovery has traditionally suffered from the fundamental challenge that learning algorithms work under the assumption of identically distributed data at train and test times, but discovery is inherently an out-of-distribution effort. Our results on large-scale learning provide a potential step to move past this dilemma, by demonstrating that GNoME models exhibit emergent out-of-distribution capabilities at scale. This includes discovery in unseen chemical spaces (for example, with more than four different elements), as well as on new downstream tasks (for example, predicting kinetic properties).

GNoME models have already found 2.2 million stable crystals with respect to previous work and enabled previously impossible modelling capabilities for materials scientists. Some open problems remain for the transition of findings in applications, including a greater understanding of phase transitions through competing polymorphs, dynamic stability arising from vibrational profiles and configurational entropies and, ultimately, synthesizability. Nevertheless, we see pretrained, general-purpose GNoME models being used as powerful tools across a diverse range of applications to fundamentally accelerate materials discovery.

## Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-023-06735-9>.

- Green, M. A., Ho-Baillie, A. & Snaith, H. J. The emergence of perovskite solar cells. *Nat. Photon.* **8**, 506–514 (2014).
- Mizushima, K., Jones, P., Wiseman, P. & Goodenough, J. B.  $Li_xCoO_2$  ( $0 < x < 1$ ): a new cathode material for batteries of high energy density. *Mater. Res. Bull.* **15**, 783–789 (1980).
- Bednorz, J. G. & Müller, K. A. Possible high  $T_c$  superconductivity in the Ba–La–Cu–O system. *Z. Phys. B Condens. Matter* **64**, 189–193 (1986).
- Ceder, G. et al. Identification of cathode materials for lithium batteries guided by first-principles calculations. *Nature* **392**, 694–696 (1998).
- Tabor, D. P. et al. Accelerating the discovery of materials for clean energy in the era of smart automation. *Nat. Rev. Mater.* **3**, 5–20 (2018).
- Liu, C. et al. Two-dimensional materials for next-generation computing technologies. *Nat. Nanotechnol.* **15**, 545–557 (2020).
- Nørskov, J. K., Bligaard, T., Rossmeisl, J. & Christensen, C. H. Towards the computational design of solid catalysts. *Nat. Chem.* **1**, 37–46 (2009).
- Greeley, J., Jaramillo, T. F., Bonde, J., Chorkendorff, I. & Nørskov, J. K. Computational high-throughput screening of electrocatalytic materials for hydrogen evolution. *Nat. Mater.* **5**, 909–913 (2006).
- Gómez-Bombarelli, R. et al. Design of efficient molecular organic light-emitting diodes by a high-throughput virtual screening and experimental approach. *Nat. Mater.* **15**, 1120–1127 (2016).
- de Leon, N. P. et al. Materials challenges and opportunities for quantum computing hardware. *Science* **372**, eabb2823 (2021).
- Wedig, A. et al. Nanoscale cation motion in TaO<sub>x</sub>, HfO<sub>x</sub>, and TiO<sub>x</sub> memristive systems. *Nat. Nanotechnol.* **11**, 67–74 (2016).
- Brown, T. et al. Language models are few-shot learners. *Adv. Neural Inf. Process. Syst.* **33**, 1877–1901 (2020).
- Dosovitskiy, A. et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR, 2021)*; <https://openreview.net/forum?id=YicbFdNTTy>
- Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).